.NET Microservices Training Curriculum

5-Day Comprehensive Technical Training Plan

Duration: 5 Days (40 Hours) **Target Framework:** ASP.NET 8 / .NET **Format:** Hands-on Workshop

Day 1 — .NET Introduction & C# Fundamentals (Part 1)

MODULE / TOPIC	COVERAGE	HANDS-ON / OUTCOME
Module 1 **Introduction to .NET 8 & C# Basics**	Overview of the .NET ecosystem, runtime, CTS/CLS, C# syntax, data types, control flow, loops, and basic error/exception handling.	Write and execute foundational C# console programs.
Module 2 **Object-Oriented Programming (OOP)**	Classes, objects, constructors, properties, inheritance, polymorphism, interfaces, and abstract classes in C#.	Build an object-oriented class library modeling domain entities.

Day 2 — C# Advanced Features & Microservices Foundations

MODULE / TOPIC	COVERAGE	HANDS-ON / OUTCOME
Module 3 **Advanced C# Language Features**	Generics, extension methods, delegates, events, and working with collections.	Implement reusable generic components and custom collections.
Module 4 **LINQ & Asynchronous Programming**	Language Integrated Query (LINQ) syntax, deferred execution, async/await patterns, and task-based asynchronous programming.	Write efficient query operations and asynchronous data fetch routines.
Module 5 **Microservices Architecture Foundations**	Monolith vs. microservices, advantages and trade-offs, bounded contexts, and the database-per-service core concept.	Define service boundaries for a sample Order Management solution.

Day 3 — Building APIs, Dependency Injection & Communication

MODULE / TOPIC	COVERAGE	HANDS-ON / OUTCOME
Module 6 **Dependency Injection & Web APIs**	Built-in Dependency Injection (DI) container, service lifetimes, controllers, DTOs, HTTP status codes, and REST principles.	Set up an ASP.NET Core project and create Product/Customer Web APIs.
Module 7 **Service-to-Service Communication**	Synchronous REST calls, gRPC overview, and asynchronous event-driven messaging basics (RabbitMQ/Kafka concepts).	Implement inter-service communication where Order service calls Product/Customer services.
Module 8 **API Versioning & Resilience**	API versioning strategies (URL/header), handling transient faults, retries, circuit breakers, and resilience with Polly.	Add a resilience policy to service-to-service calls using Polly.

Day 4 — Security, Testing, Logging & Observability

MODULE / TOPIC	COVERAGE	HANDS-ON / OUTCOME
Module 9 **Securing Microservices**	JWT authentication, claims, role-based authorization, and securing API endpoints.	Secure Order and Product APIs using JWT tokens.
Module 10 **Testing Microservices**	Unit testing fundamentals, mocking dependencies, and integration testing concepts for APIs.	Write unit and integration tests for service behavior.
Module 11 **Logging & Observability**	Structured logging with Serilog, request correlation identifiers, health checks, and basic metrics (Prometheus/Grafana overview).	Configure structured logging and expose health check endpoints.

Day 5 — Cloud Deployment Fundamentals

MODULE / TOPIC	COVERAGE	HANDS-ON / OUTCOME
Module 12 **Azure Key Vault Basics**	Purpose of Azure Key Vault, managing keys and secrets securely, and externalizing application configuration.	Store a sample connection string/secret and configure microservices to retrieve it securely.
Module 13 **Azure App Service Deployment**	App Service concepts, application settings, environment configuration, and deploying .NET Web APIs.	Deploy a .NET microservice to Azure App Service and validate end-to-end API connectivity.